

Unit 2 Review Packet

Data



How to use this packet

Match the vocabulary, then work the mixed review questions. These are the exact words and question types you'll see on the unit test and the AP exam.

Vocab Match 1

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|----------------------------|--|
| ___ 1. Bit | A. real numbers are approximated because only a fixed number of bits is used |
| ___ 2. Analog data | B. compression where no information is lost; the original can be restored |
| ___ 3. Sampling | C. compression that permanently removes some data |
| ___ 4. Overflow error | D. data with values that change smoothly over time (pitch, volume, position) |
| ___ 5. Roundoff error | E. repeated patterns in data that compression takes advantage of |
| ___ 6. Data compression | F. approximating analog data digitally by measuring values at discrete intervals |
| ___ 7. Lossless | G. short for binary digit; the smallest unit of data, 0 or 1 |
| ___ 8. Lossy | H. occurs when a number is too large for the fixed number of bits used to store it |
| ___ 9. Run-length encoding | I. reducing the number of bits needed to represent data |
| ___ 10. Redundancy | J. replaces runs of repeated values with a count and the value |

Vocab Match 2

- | | |
|---------------------------|--|
| ___ 1. Information | A. choosing a subset of data that meets certain criteria |
| ___ 2. Correlation | B. arranging data into order (alphabetical, numerical, by date) |
| ___ 3. Data cleaning | C. a system's capacity to change in size and scale to meet new demands |
| ___ 4. Scalability | D. making data uniform without changing its meaning (e.g., fixing spellings/abbreviations) |
| ___ 5. Bias in data | E. changing every element in a data set in the same way (e.g., °F → °C) |
| ___ 6. Filtering | F. caused by the type or source of data; NOT fixed by just collecting more data |
| ___ 7. Sorting | G. repeating steps for every element in a data set |
| ___ 8. Data visualization | H. the collection of facts and patterns extracted from data |
| ___ 9. Transforming | I. a graphical representation that reveals patterns |
| ___ 10. Iteration | J. a relationship between two variables; one changes as the other changes |

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Mixed Review

1. A data set has: student ID, grade level, number of absences, and GPA. Which TWO questions can be answered? (*Select two.*)

- A. Is there a relationship between absences and GPA?
- B. Do students who live farther from school miss more days?
- C. Which grade level has the highest average number of absences?
- D. Why do students miss school?

2. Which situation most requires **lossless** compression?

- A. Streaming a movie
- B. A video call with friends
- C. Posting a profile picture
- D. Sending a spreadsheet of student grades

3. Which visualization best shows how average global temperature changed from 1950 to 2020?

- A. Line graph
- B. Pie chart
- C. A single table cell
- D. Word cloud

4. Which list of binary numbers is in order from **least to greatest**?

- A. 1001, 0111, 1100, 1010
- B. 0111, 1010, 1001, 1100
- C. 1100, 1010, 1001, 0110
- D. 0110, 1001, 1010, 1100

5. A music service stores for each song: title, artist, genre, length, and number of plays. Which question **CANNOT** be answered using only this data?

- A. Which genre has the most total plays?
- B. What is the average song length for each artist?
- C. Which age group listens to the most songs?
- D. Which artist has the most songs in the library?

6. A game stores a player's score using **4 bits** (values 0–15). The player has 14 points and earns 3 more. What is most likely to happen?

- A. The score correctly shows 17
- B. An overflow error occurs
- C. A roundoff error occurs
- D. A syntax error occurs

7. Which is an example of **metadata** for a digital photo?

- A. The colors of the pixels
- B. The date the photo was taken
- C. The people in the photo
- D. The brightness of the image

8. A data set lists states as "CA", "Calif.", and "California". Which process should be applied?

- A. Data compression
- B. Encryption
- C. Data cleaning
- D. Sampling

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9. A school stores locker numbers using 6 bits. It now needs 100 lockers. Which is true?

- A. 6 bits represents only 64 values; at least 7 bits are needed
- B. 6 bits is enough because $6 \times 20 = 120$
- C. 100 bits are needed, one per locker
- D. 8 bits are the minimum because a byte is 8 bits

11. A photo app wants to minimize upload time and storage. Users don't notice small quality loss. Which is the best choice?

- A. Lossy, because size and speed matter more than perfect quality
- B. Lossless, because it keeps every detail
- C. No compression
- D. Lossless, because it always makes smaller files

13. What is the decimal value of binary **1101**?

- A. 13
- B. 11
- C. 14
- D. 1101

15. Which statement about lossy compression is true?

- A. The original can always be perfectly restored
- B. It only works on text files
- C. It usually achieves a greater reduction in size than lossless
- D. It never changes the quality of an image

10. Researchers find that towns with more libraries tend to have more crimes. Which conclusion is most appropriate?

- A. Libraries cause crime
- B. The variables are correlated; another factor (like population) may explain both
- C. Closing libraries would lower crime
- D. The data must be wrong

12. A library database stores: title, author, genre, year published, and times checked out. Which steps find the **most-checked-out mystery book published after 2000**?

- A. Filter genre = mystery; filter year > 2000; sort by checkouts (high→low); take the first
- B. Sort by checkouts; take the first; then filter genre = mystery
- C. Filter year > 2000; sort by title; take the first
- D. Filter genre = mystery; sort by year; take the last

14. Which TWO statements are true? (*Select two.*)

- A. Lossless algorithms can guarantee reconstruction of the original data.
- B. Lossy compression always produces a smaller file than lossless for every file.
- C. The amount of compression depends on the redundancy in the data.
- D. Compressing a file twice with a lossless algorithm always makes it smaller again.

16. Which is an example of **filtering** data?

- A. Converting prices from dollars to euros
- B. Arranging products from cheapest to most expensive
- C. Showing only products that cost under \$20
- D. Adding up the price of every product

Name: _____

Class: _____

Date: _____

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Challenge Questions

Challenge 2.1

IPv4 addresses use **32 bits**. IPv6 addresses use **128 bits**. How many times more addresses does IPv6 allow than IPv4? Show your reasoning with powers of 2, and explain why this change was needed.

Challenge 2.2

A hospital wants to save storage space on its X-ray images. An administrator suggests **lossy** compression because it saves the most space. Argue for or against this plan, explaining the trade-off.

Challenge 2.3

A company tests a voice assistant using recordings from 10,000 employees at its headquarters. Engineers suggest collecting 50,000 more recordings from the *same* office to reduce bias. Explain why this will not fix the problem and what they should do instead.

Challenge 2.4

A city uses step-count data from a fitness smartwatch app to decide where to build new parks. Identify a limitation of this data set and explain how it could lead to an unfair decision.
